

# SATQUERY AI — REMOTE SENSING ANALYSIS REPORT

ISRO Problem Statement 26167 • Mission ID: 47cbdb67 • Generated: 14:43:47.041Z

|                        |                                               |
|------------------------|-----------------------------------------------|
| Natural Query:         | Use both images to identify built-up regions. |
| Classified Task:       | OPTICAL_SAR_ANALYSIS                          |
| Calibrated Confidence: | 92% (Multi-Sensor Cross-Modal Concordance)    |
| Specialist Models:     | satquery-optical-sar-fusion-v1                |
| Total Latency:         | 301.7 ms                                      |

## GROUNDING AI INTERPRETATION

|                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Optical-SAR synergistic analysis identified 11.33% built-up infrastructure. SAR microwave backscatter effectively validated structural footprints while bypassing optical cloud/atmospheric shadows. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## INPUT SENSOR METADATA

| Filename             | Modality | Dimensions | Bands | GeoTIFF CRS |
|----------------------|----------|------------|-------|-------------|
| isro_sdsc_optical.ti | OPTICAL  | 512x512    | 3     | EPSG:4326   |
| isro_sdsc_sar.tif    | SAR      | 512x512    | 1     | EPSG:4326   |

## OBSERVABLE AGENTIC EXECUTION TRACE

| # | Phase / Event                                     | Observable Details                                                                                         | Time (m) |
|---|---------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------|
| 1 | Query Ingestion                                   | Received natural language query: 'Use both images to identify built-up regions.' with 2 uploaded image(s). | 0.0      |
| 2 | Input Validation                                  | Verified 2 image file(s). Format and size limits satisfied.                                                | 0.0      |
| 3 | Raster Metadata Extraction: isro_sdsc_optical.tif | Extracted dimensions (512x512), 3 band(s). CRS: EPSG:4326                                                  | 66.4     |
| 4 | Raster Metadata Extraction: isro_sdsc_sar.tif     | Extracted dimensions (512x512), 1 band(s). CRS: EPSG:4326                                                  | 28.1     |
| 5 | Modality Identification                           | Detected sensor modalities: ['OPTICAL', 'SAR'].                                                            | 0.0      |
| 6 | Query Classification → OPTICAL_SAR_ANALYSIS       | Dual-sensor pair (Optical + SAR) detected -> Routed to Cross-Modal Feature Fusion Specialist.              | 0.0      |
| 7 | Model Dispatch                                    | Dispatched task to specialist: 'satquery-optical-sar-fusion-v1' on device 'cpu'.                           | 0.0      |
| 8 | Specialist Execution Completed                    | Inference finalized in 206.8ms. Evidence artifacts compiled.                                               | 206.8    |
| 9 | Result Fusion & Calibration                       | Derived grounded response with calibrated confidence 92% (Multi-Sensor Cross-Modal Concordance).           | 0.0      |

